Session 3: Object Detection Techniques for Earth Observation

From Faster R-CNN to YOLO/DETR: Counting and Localizing Objects

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Session Overview

Duration: 1.5 hours **Type:** Theory + Discussion **Goal:** Master object detection for Earth Observation

You will learn:

- What object detection is and when to use it
- Two-stage vs single-stage vs transformer-based detectors
- Key concepts: anchor boxes, NMS, IoU, mAP
- EO applications: ships, vehicles, buildings, infrastructure
- Architecture selection for Philippine use cases

Prerequisites:

- Day 2 CNNs and Transfer Learning
- Basic understanding of image classification
- Familiarity with Python and deep learning

Resources:

- Hands-on lab in Session 4
- Reference PDF: “Day 3, Session 3_ Object Detection for Earth Observation Imagery.pdf”

Part 1: What is Object Detection?

Definition & Core Concept

Object detection combines two fundamental computer vision tasks:

1. **Classification:** What objects are present in the image?
2. **Localization:** Where exactly are they located?

For each detected object, the model outputs:

- **Bounding box coordinates:** (x, y, width, height)
- **Class label:** Object type (ship, vehicle, building)
- **Confidence score:** Detection probability (0 to 1)

Task Comparison Matrix

Aspect	Classification	Object Detection	Semantic Segmentation
Question	What's in this image?	Where are objects?	Which pixels are what?
Output	Single label	Bounding boxes + labels	Pixel-wise mask
Granularity	Image-level	Object-level	Pixel-level
Location Info	None	Approximate (boxes)	Precise (pixels)
Count Objects	No	Yes ✓	Possible
Speed	Fastest	Moderate	Slowest
Example Use	"Contains buildings"	"10 buildings at coords"	"Building footprints mapped"

When to Use Each Task?

Choose Object Detection when:

- Need to **count individual instances** (ships, vehicles, aircraft)
- Bounding box approximation is sufficient
- Real-time or near-real-time processing required
- Objects are well-separated and distinct

EO Examples:

- Ship counting in maritime zones
- Vehicle detection for traffic analysis
- Aircraft monitoring at airports

Choose Semantic Segmentation when:

- Need **precise pixel-level boundaries** (flood extent, land parcels)
- Accurate area calculations are critical
- Objects have complex shapes not suited to rectangles
- Can afford slower processing time

EO Examples:

- Mapping typhoon flood extent
- Agricultural field delineation
- Building footprint mapping

Philippine Example

Detection vs Segmentation

Counting ships in Manila Bay → Object Detection

Mapping typhoon flood extent → Semantic Segmentation

Part 2: Architecture Families

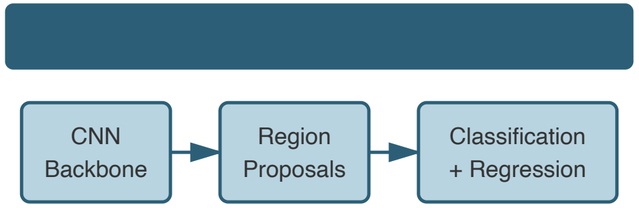
Evolution of Object Detection

Object detection has evolved through three main architectural paradigms:

1. **Two-Stage Detectors** (2015-2017): Separate proposal + classification
2. **Single-Stage Detectors** (2016-Present): Direct prediction in one pass
3. **Transformer-Based** (2020-Present): Set prediction without anchors

Architecture Comparison

Object Detection Architecture Families



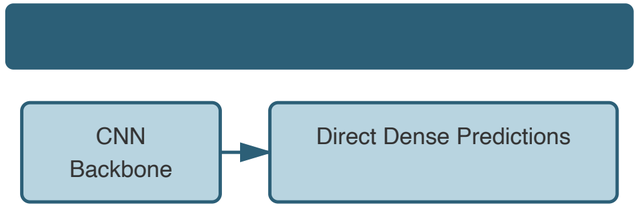
Pros:

✓ High accuracy

✓ Good for small objects

Cons:

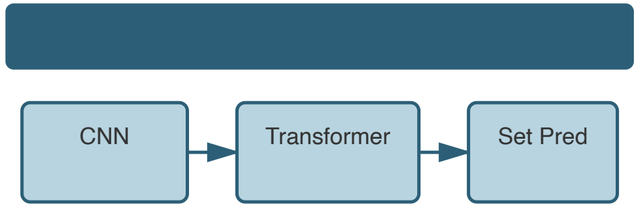
✗ Slower (~200ms/image)



Pros:

✓ Fast (45+ FPS)

✓ Real-time capable



Pros:

✓ No anchors

Two-Stage Detectors: Faster R-CNN

Pipeline (4 stages):

1. **CNN Backbone** (e.g., ResNet)
   - Extracts feature map from input image
2. **Region Proposal Network (RPN)**
   - Generates object proposals
3. **ROI Pooling**
   - Extracts features for each proposal
4. **Classification + Box Regression**
   - Predicts class and refines coordinates

Faster R-CNN Performance

Performance Metrics

- ~200ms per image on GPU
- High accuracy (mAP ~40% on COCO)
- Excellent for small objects

Pros

- ✓ High accuracy
- ✓ Good for small objects
- ✓ Handles scale variation well

Cons

- ✗ Slower inference
- ✗ More complex architecture

Best for

Precision-critical EO applications:

- Detailed infrastructure surveys
- Small object inventories
- High-precision damage assessment

Key Innovation

RPN replaces slow Selective Search with learned proposal generation

Single-Stage Detectors: YOLO

Core Concept: Reframe detection as regression—predict boxes and classes in single forward pass

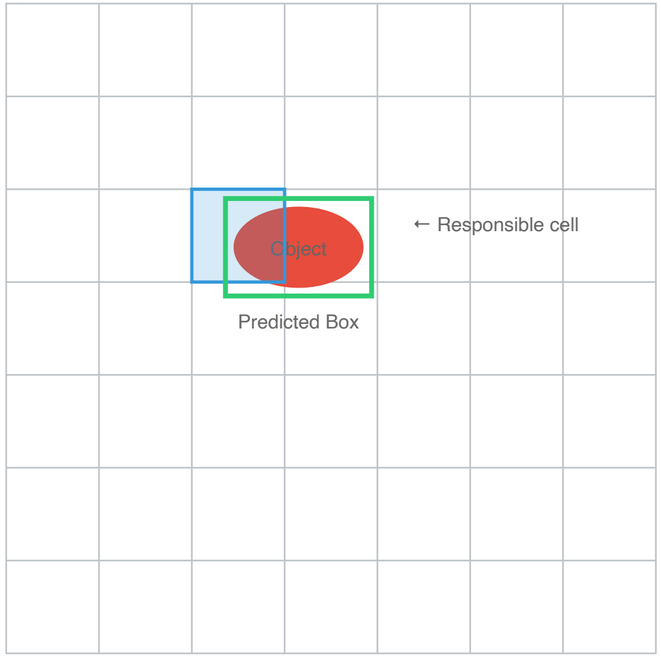
- Divide image into S×S grid (e.g., 7×7 or 13×13)
- Each grid cell predicts B bounding boxes + confidence
- Each cell predicts C class probabilities
- Output tensor: S × S × (B × 5 + C)

YOLO: Grid-Based Detection

YOLO: You Only Look Once

Grid-Based Object Detection

Input Image divided into S×S Grid (e.g., 7×7)



How YOLO Works:

1. Divide image into S×S grid
   - Each cell: responsible for objects whose center falls in the cell
2. Each cell predicts B boxes
   - Box: (x, y, w, h, confidence)
   - Confidence: P(Object) × IoU
3. Each cell predicts C classes
   - Class probabilities for object
4. Output tensor shape
   - S × S × (B × 5 + C)
   - Example: 7 × 7 × (2 × 5 + 20) = 7 × 7 × 30

Key Advantage:

✓ Single forward pass → FAST (45+ FPS)

YOLO Evolution:

YOLOv1-v3	YOLOv5	YOLOv7	YOLOv8
2016-2018	PyTorch, practical (2020)	Speed optimizations (2022)	Anchor-free, best (2023)

YOLO Evolution Timeline

Version	Year	Key Features	Notes
YOLOv1-v3	2016-2018	Original development	Grid-based, anchor boxes
YOLOv4	2020	Bag of tricks	CSPDarknet, Mish activation
YOLOv5	2020	PyTorch implementation	Very practical, widely used
YOLOv7	2022	Speed optimizations	E-ELAN, compound scaling
YOLOv8	2023	Anchor-free (Ultralytics)	Widely used baseline
YOLOv9 / v10	2024	PGI/GELAN; NMS-free training	Research releases
YOLO11	2024	Ultralytics; OBB support	Oriented boxes for aerial imagery
YOLO26	Jan 2026	Ultralytics; end-to-end, NMS-free	Current Ultralytics family

Performance: real-time on a GPU; COCO mAP competitive with two-stage methods

For Philippine EO

A current Ultralytics YOLO (YOLO11/YOLO26; OBB variant for ships) is a practical default for real-time monitoring. Ultralytics is AGPL-3.0 — check licensing for operational use.

Single-Stage: SSD (Single Shot MultiBox)

Key Innovation

Multi-scale feature maps for detecting objects at different scales

Architecture:

- Base network (VGG or ResNet)
- Multiple convolutional feature layers at different resolutions
- Each layer predicts objects at specific scales
- Combines predictions from all layers

Advantage over YOLO: Better at detecting small objects

Use Cases

- When objects vary significantly in size
- Small object detection (vehicles, small ships)
- Balance between speed and accuracy

Performance

- 59 FPS on VGG16 backbone
- Good accuracy on small objects
- mAP competitive with Faster R-CNN

Best for: Applications requiring both speed and small object detection

Transformer-Based: DETR

Detection Transformer (DETR)

- End-to-end object detection with transformers
- Set prediction via bipartite matching
- No anchors, no NMS needed
- Elegant architecture

How it works:

1. CNN backbone extracts features
2. Transformer encoder processes features
3. Transformer decoder predicts set of objects
4. Bipartite matching for training

Pros

- ✓ No hand-crafted components (anchors, NMS)
- ✓ Elegant end-to-end design
- ✓ Good for complex scenes

Cons

- ✗ Requires more training data
- ✗ Longer convergence time
- Real-time variants (RT-DETR, RF-DETR) now close the speed gap with YOLO
- ✗ Slower than YOLO/SSD

Best for: Research applications, complex multi-object scenes when data is plentiful

Part 3: Key Concepts

Anchor Boxes

Purpose

Provide reference boxes at various scales and aspect ratios that the model adjusts to fit actual objects

How They Work

- Predefined boxes placed at each spatial location
- Model predicts **offsets** (adjustments) to anchors
- Example: 9 anchors per location (3 scales × 3 aspect ratios)

Example Anchors:

- Scales: 128^2, 256^2, 512^2 pixels
- Aspect ratios: 1:1, 1:2, 2:1

Modern Alternatives

- **Anchor-free methods** (FCOS, CenterNet, YOLOv8+)
- Predict object centers and dimensions directly
- Simpler, fewer hyperparameters
- Better generalization

Trade-offs

- Anchor-based: More established, predictable
- Anchor-free: Simpler, more flexible

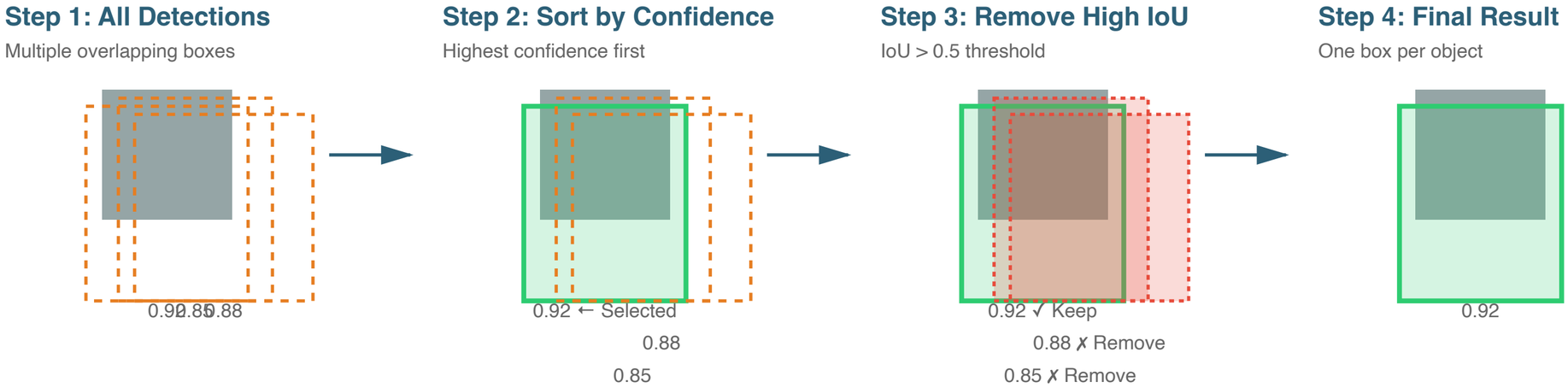
Non-Maximum Suppression (NMS)

Problem: Detectors generate multiple overlapping boxes for the same object

Solution: Remove duplicate detections through NMS algorithm

NMS Process Visualization

Non-Maximum Suppression (NMS) Process



NMS Algorithm:

1. Sort all detections by confidence score (highest first)
2. Select box with highest confidence → add to final detections
3. Remove all boxes with IoU > threshold (e.g., 0.5) with selected box
4. Repeat until no boxes remain

Threshold Trade-off:

• Lower (0.3): Stricter, may remove valid overlapping objects

• Higher (0.7): Looser, may keep duplicate detections

• Typical: 0.4-0.5

NMS Algorithm Steps

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Intersection over Union (IoU)

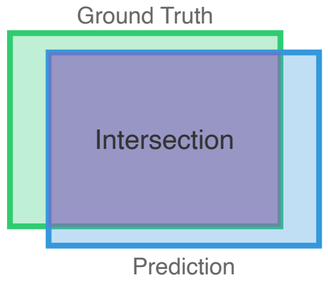
Definition: Measure of overlap between two bounding boxes

$$\text{IoU} = \frac{\text{Area of Intersection}}{\text{Area of Union}}$$

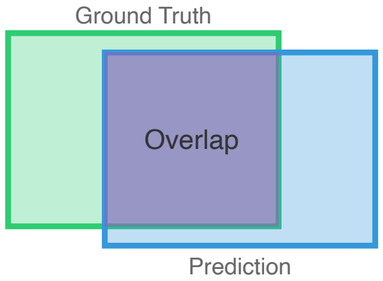
IoU Visualization

Intersection over Union (IoU)

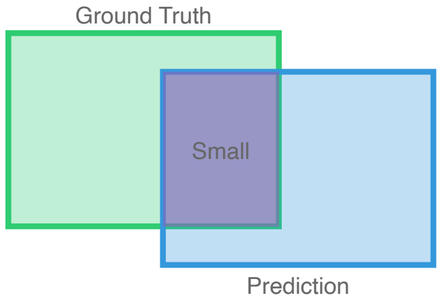
Good Detection (IoU = 0.75)



Medium Detection (IoU = 0.50)



Poor Detection (IoU = 0.20)



IoU Formula

IoU = Area of Intersection / Area of Union

Three Critical Uses:

1. **Training:** Match predictions to ground truth (IoU > 0.5 = positive match)
2. **NMS:** Remove overlapping duplicate detections

Interpretation:

- IoU = 1.0: Perfect overlap
- IoU = 0.5: Standard threshold

IoU: Three Critical Uses

1. **Training:** Match predictions to ground truth (IoU > 0.5 = positive match)
2. **NMS:** Remove overlapping duplicate detections
3. **Evaluation:** Determine if detection is correct

Interpretation:

- IoU = 1.0: Perfect overlap
- IoU = 0.5: Standard threshold for “correct” detection
- IoU = 0.0: No overlap

Mean Average Precision (mAP)

Primary Evaluation Metric

Calculation Steps:

1. For each class:
   - Sort detections by confidence
   - Compute precision-recall curve
   - Calculate Average Precision (AP) = area under curve
2. mAP = average AP across all classes

Common Variants:

- **mAP@0.5:** IoU threshold = 0.5 (PASCAL VOC)
- **mAP@0.75:** Stricter threshold
- **mAP@[0.5:0.95]:** Average over range (COCO)

Example Interpretation

- mAP@0.5 = 0.65 → Model achieves 65% average precision at IoU ≥ 0.5
- Higher is better (max = 1.0)

Why Multiple Thresholds?

- mAP@0.5: Lenient, accepts rough localizations
- mAP@0.75: Strict, requires precise boxes
- mAP@[0.5:0.95]: Comprehensive evaluation

For EO Applications:

- Use mAP@0.5 for quick evaluation
- Use mAP@0.75 for precision-critical tasks

Part 4: EO Applications

Application 1: Maritime Surveillance

Philippine Context: Maritime Security

Use Case: Monitor Exclusive Economic Zone (EEZ), detect illegal fishing, track vessel movements

Data Sources:

- Sentinel-1 SAR (all-weather, day-night)
- Sentinel-2 optical (higher resolution)
- High-resolution commercial imagery

Challenges:

- Small objects (10-30 pixels)
- Wave patterns (false positives)
- Scale variation (boats to cargo ships)
- No fixed orientation

Recommended Approach

- **Real-time monitoring:** YOLO (v8+, OBB) for speed
- **Precision inventory:** Faster R-CNN for accuracy
- **Multi-scale training** essential

Indicative Performance (varies with data)

- YOLO: faster, real-time on GPU
- Faster R-CNN: often slightly more accurate on small ships, several× slower

Key Considerations

- SAR requires different preprocessing
- Wave patterns need careful handling
- Multi-temporal analysis helps distinguish static objects

Ship Detection Benefits

- **Automated monitoring** of vast maritime zones
- **Illegal fishing detection** in protected areas
- **Traffic analysis** in shipping lanes
- **Search and rescue** operations support
- **Maritime security** enforcement

> **Session 4 Preview**
>
> Next session’s hands-on lab: fine-tune a pre-trained detector, framed by a Metro Manila use case.

Application 2: Urban Monitoring

Vehicle Detection

Use Case: Traffic flow, parking, planning

- **Data:** VHR satellites (<1m resolution)
- **Challenges:** Very small (5-10 pixels), dense
- **Approach:** Faster R-CNN + FPN

Indicative Performance:

- Vehicles need sub-metre imagery; mAP varies widely by dataset
- Better in sparse areas

Building Detection

Use Case: Asset inventory, informal settlement mapping, damage assessment

- **Data:** Sentinel-2 (10m) or VHR
- **Challenges:** Size variation, shadows
- **Approach:** YOLO (v8+) or Faster R-CNN

Philippine Application:

- Informal settlement monitoring (Metro Manila)
- Disaster damage assessment (post-typhoon)
- Urban growth tracking

Application 3: Infrastructure Monitoring

Oil Tank Detection

- Distinctive circular shape aids detection
- High reflectance in optical imagery
- Strategic infrastructure monitoring
- Regular size distribution

Approach: Faster R-CNN or circular Hough + CNN

Performance: typically among the easiest EO objects (distinct circular shape)

Aircraft Detection

- Airport activity monitoring
- Medium-sized objects (20-50 pixels)
- High contrast against runways
- Fewer orientation variations

Approach: YOLO series for real-time

Performance: high contrast makes aircraft comparatively easy in sub-metre imagery

Use Cases: Airport traffic analysis, capacity planning

Part 5: EO-Specific Challenges

Challenge 1: Small Object Detection

The Problem

Many EO objects are tiny relative to image size:

- Ships: 10-30 pixels in Sentinel-1/2
- Vehicles: 5-10 pixels in satellite imagery

Why Difficult

- Feature maps downsample spatial resolution
- Small objects lose critical information
- Insufficient features for classification
- Standard anchors may not cover small scales

Solutions

1. **Multi-scale feature pyramids** (FPN architecture)
2. **Higher-resolution input images**
3. **Tile large images** into smaller patches
4. **Multi-scale training** with different input sizes
5. **Specialized architectures** for small objects
6. **Data augmentation** focusing on scale

Best Practices

- Use FPN or PANet architectures
- Train with tiles (512×512 or 1024×1024)
- Include small-scale anchors

Challenge 2: Scale & Orientation Variation

Scale Variation

Same object type at vastly different scales:

- Small fishing boat vs. cargo ship (10× difference)
- Compact car vs. truck
- Single-story vs. high-rise building

Orientation

Objects at any angle in overhead imagery:

- Ships at various headings
- No canonical viewpoint
- 360° rotation possible

Solutions

1. **Multi-scale anchor boxes** covering expected ranges
2. **Rotation-invariant features** or augmentation
3. **Oriented bounding boxes** (OBB) instead of axis-aligned
4. **Multi-scale training** protocol
5. **Rotation augmentation** during training

Implementation

- YOLO (OBB variant) with rotation augmentation
- Rotated Region CNN (R2CNN) for OBB
- Data augmentation: 0°, 90°, 180°, 270°

Challenge 3: Complex Backgrounds

The Problem

Satellite imagery contains many distractors:

- Urban clutter (shadows, roads, rooftops)
- Wave patterns mimicking ships
- Terrain texture variations
- Seasonal appearance changes

Impact

- False positives increase
- Model confidence decreases
- Precision suffers

Solutions

1. **Contextual information** (use surrounding region)
2. **Stronger backbone networks** (ResNet-101, EfficientNet)
3. **Hard negative mining** during training
4. **Domain-specific preprocessing**
5. **Attention mechanisms** to focus on features

Best Practices

- Include diverse backgrounds in training
- Use hard negative mining
- Apply domain-specific filters
- Consider contextual features

Challenge 4: Limited Labeled Data

The Problem

Annotating EO imagery is expensive and time-consuming:

- Requires domain expertise
- Large images mean many objects
- Ground truth often unavailable
- Manual annotation costly

Impact

- Model overfitting
- Poor generalization
- Limited object type coverage

Solutions

1. **Transfer learning** from COCO or ImageNet
2. **Self-supervised pre-training** on unlabeled EO data
3. **Semi-supervised learning** approaches
4. **Active learning** for informative samples
5. **Synthetic data generation**
6. **Data augmentation** to increase dataset size

Philippine Context

- Use pre-trained models from similar domains
- Collaborate with government agencies
- Leverage existing datasets (xView, DOTA)

Challenge 5: Atmospheric & Quality Issues

Unique EO Challenges

- Cloud cover and shadows
- Haze and aerosol scattering
- Seasonal illumination variations
- Sensor noise and artifacts
- Variable imaging conditions

Impact

- Reduced image quality
- Inconsistent appearance
- Lower detection confidence
- Temporal variations

Solutions

1. **Robust preprocessing** (atmospheric correction, normalization)
2. **SAR data** as weather-independent alternative
3. **Multi-temporal fusion** to handle clouds
4. **Domain adaptation** techniques
5. **Augmentation** simulating atmospheric effects

Best Practices

- Use Sentinel-1 SAR for all-weather capability
- Apply atmospheric correction
- Normalize across scenes
- Train on diverse conditions

Part 6: Architecture Selection

Decision Framework: Two-Stage Detectors

Use Faster R-CNN When:

- ✓ **Accuracy is paramount** over speed
- ✓ Detecting **small objects** (ships, vehicles)
- ✓ Objects have **high scale variation**
- ✓ Can afford **longer inference time**
- ✓ Have **good quality labeled training data**

Example EO Applications:

- Detailed infrastructure surveys
- Small object inventories (ship counting)
- High-precision damage assessment
- Building footprint extraction

Decision Framework: Single-Stage Detectors

Use YOLO or SSD When:

- ✓ **Speed is critical** (real-time monitoring)
- ✓ Objects are **medium-to-large** size
- ✓ Can accept **slightly lower accuracy**
- ✓ Need to **process many images** quickly
- ✓ **Resource-constrained** deployment

Example EO Applications:

- Real-time maritime surveillance
- Rapid disaster response (initial assessment)
- Large-area scanning for anomalies
- Drone video analysis

Philippine EO Application Guide

Application	Recommended Architecture	Key Reasoning
Ship Detection (EEZ)	YOLO-OBB or Faster R-CNN	Balance speed vs accuracy
Vehicle Counting	Faster R-CNN + FPN	Very small objects
Building Detection	YOLO (v8+)	Good balance
Aircraft Monitoring	YOLO series	Real-time, medium objects
Oil Tank Inventory	Faster R-CNN	Precision matters
Disaster Assessment	YOLO (v8+)	Rapid large-area scans
Informal Settlement	YOLO or Faster R-CNN	Speed vs detail trade-off

Decision Framework: Transformer-Based

Use DETR When:

- ✓ Have **large training dataset**
- ✓ Want **elegant architecture** without heuristics
- ✓ Can afford **longer training time**
- ✓ Working on **research/cutting-edge** applications
- ✓ **Complex scene understanding** matters

Considerations:

- Requires more data than CNN-based methods
- Longer training convergence
- May not be faster than optimized YOLO
- Good for eliminating manual tuning

Current Status:

- Still evolving, less mature than YOLO/Faster R-CNN
- Better suited for research than production
- Consider Deformable DETR for better performance

Part 7: Resources & Summary

Foundational Papers

Two-Stage Detectors

- Girshick et al. (2014) - R-CNN: Rich feature hierarchies
- Ren et al. (2015) - Faster R-CNN with RPN

Single-Stage Detectors

- Redmon et al. (2016) - YOLO: Unified detection
- Liu et al. (2016) - SSD: Single Shot MultiBox
- Lin et al. (2017) - RetinaNet with Focal Loss

Transformer-Based

- Carion et al. (2020) - DETR: End-to-End Detection

EO-Specific

- Ship detection surveys
- Building detection in remote sensing
- Small object detection in aerial imagery

Frameworks and Datasets

Popular Implementations

- Ultralytics YOLO - YOLOv8/11/26, user-friendly (AGPL-3.0)
- Detectron2 - Facebook AI framework
- MMDetection - Comprehensive toolbox

Pre-trained Models

- torchvision detection models (COCO weights)
- Ultralytics OBB models (DOTA-pretrained)
- TF2 Detection Model Zoo (legacy)

EO-Specific Datasets

- **DOTA:** Large-scale aerial images
- **DIOR:** Optical remote sensing
- **xView:** Largest overhead imagery dataset

Philippine EO Resources

- PhilSA: Maritime surveillance
- NAMRIA: Infrastructure mapping
- DOST-ASTI: DATOS disaster response

Key Takeaways

Object Detection Fundamentals

- Combines classification + localization with bounding boxes
- Outputs: box coordinates, class, confidence
- Distinct from classification and segmentation
- Primary use: counting and locating instances

Architecture Categories

- **Two-stage:** Accurate, slower, good for small objects
- **Single-stage:** Fast, real-time, competitive accuracy
- **Transformer:** Elegant, data-hungry, research-focused

Key Concepts

- **Anchor boxes** provide reference scales
- **NMS** removes duplicate detections
- **IoU** measures box overlap
- **mAP** is the standard evaluation metric

EO-Specific Challenges

- Small objects, scale variation, complex backgrounds
- Atmospheric effects, limited labeled data
- Solutions: multi-scale features, transfer learning, augmentation

Next: Session 4 Hands-on Lab

What You'll Do:

1. Explore Sentinel-2 imagery of Metro Manila (what 10 m can show)
2. Load a COCO-pretrained detector (torchvision / Ultralytics YOLO)
3. Walk through fine-tuning on a small annotated dataset
4. Evaluate performance using mAP metric
5. Visualize detected bounding boxes
6. Export results for GIS integration

Case Study: Building Detection in Metro Manila

- **Location:** Metro Manila (Quezon City or Pasig River)
- **Relevance:** Urban monitoring, disaster vulnerability
- **Format:** guided walkthrough (no dedicated notebook in this edition)
- **Approach:** Transfer learning with pre-trained detector

Expected Results: a working fine-tuning recipe and realistic expectations for 10 m data

Discussion Questions

Before moving to the hands-on session, consider:

1. **For your EO applications**, would you prioritize speed (YOLO) or accuracy (Faster R-CNN)? Why?
2. **How would you handle** limited labeled training data for Philippine-specific object types?
3. **What validation strategy** would ensure your model generalizes across different Philippine regions?
4. **How might you combine** object detection with other techniques for disaster damage assessment?

Thank You!

Questions?

Next: Session 4 - Hands-on Object Detection Lab

Contact: Dimitrios Kasampalis, Anefelo